

NITROGEN PURGING AND ISOLATION PROCEDURE - NHT REACTOR LOOP

Document No.	NHT-SOP-114	Revision	Rev 3
Unit	Unit 24 - Naphtha Hydrotreater (NHT)	Date	15-Jan-2026
Revision Status	CURRENT - APPROVED FOR FIELD USE		
Effective Date	15-Jan-2026		
Supersedes	NHT-SOP-114 Rev 2 dated 22-Aug-2024		
Next Review Due	15-Jan-2029		

4. NITROGEN PURGING REQUIREMENTS (Rev 3)

Parameter	Rev 3 Requirement
Minimum nitrogen purge duration	4 hours
Oxygen acceptance criterion	less than 5.0 vol%
Number of blind (spade) points	6
LEL criterion	less than 1.0 % LEL
Gas test frequency during entry	Every 2 hours

Changes from Rev 2: purge duration increased from 2 hours to 4 hours; oxygen criterion tightened from 8.0 vol% to 5.0 vol%; two additional blind points added (BL-05, BL-06); an LEL criterion has been introduced; gas test frequency during entry increased.

5. BLIND LIST (Rev 3)

BL-01 Reactor feed inlet | BL-02 Reactor effluent outlet | BL-03 Hydrogen make-up line | BL-04 Flare header tie-in | BL-05 Recycle gas return | BL-06 Wash water injection

6. ISOLATION OF V-2405

Positive isolation of V-2405 for entry requires closure of the two hand valves HV-2431 (inlet) and HV-2432 (outlet), followed by insertion of blinds BL-01 and BL-02. PSV-2418 is a relief device protecting V-2405 and shall NOT be closed, gagged or isolated as part of any isolation activity while the vessel remains in a pressurisable condition.

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